Week 5 — Reading Pack

Curated excerpts from industry white papers, the syllabus readings, the PMSA items, and the industry sources behind tonight's vocabulary — organized by learning objective

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Required reading before Week 6 · ≈35 minutes (plus Athey & Imbens §1–3, ≈40 minutes)

Assigned alongside: Athey & Imbens (2016), "Recursive Partitioning for Heterogeneous Causal Effects," PNAS — §1–3; the Veeva / Salesforce Marketing Cloud NBA orchestration papers (links file); the NBA Field Guide (reference).

Part A sets the stakes in the industry's words (verbatim, with source). Part B: what an NBA engine is for. Part C: the data layer it stands on; C2: the industry's evidence on saturation. Part D: patient analytics and governance. Part E: the syllabus reading and the PMSA items. Part F: the sources behind tonight's vocabulary. Answer the reflection questions; bring one to Week 6.

Learning objectives for Week 5 (from the syllabus)

1. Frame Next Best Action as a contextual bandit / uplift problem rather than a classification problem.
2. Build an HCP affinity model and a simple NBA scoring layer.
3. Discuss how to manage physician fatigue, frequency caps, and channel substitution.

Key concepts: uplift modeling (T-learner, X-learner, causal forest) vs. response modeling; contextual bandits and offline policy evaluation; orchestration — triggers, suppression rules, and the role of the CDP.

A. Why the Omnichannel Lead's question is hard (objective 1)

Two excerpts from IQVIA's 2025 survey of commercial leaders. The first says omnichannel targeting and personalization are the highest-priority, lowest-adoption use cases in sales and marketing. The second names what comes after adoption — orchestration — which is tonight's subject.

> *Sales and Marketing: High priority, low adoption — Sales and marketing are rated highest in strategic importance, yet low current average adoption. Despite the promise of AI to drive field productivity, omnichannel targeting, and content personalization, many organizations have not yet industrialized these tools. The barriers are less about ambition and more about scaling enablement across decentralized teams.*
>
> — IQVIA, "AI in Life Sciences Commercialization" (2025 survey of 107 senior commercial leaders), AI impact by commercial function

> *As AI use cases expand and adoption deepens, the next challenge is orchestration: aligning AI with business KPIs, integrating it across functions, and continuously optimizing for impact. Successful organizations will approach AI as an enterprise capability that is intelligent, integrated, and built to scale.*
>
> — IQVIA, "AI in Life Sciences Commercialization," Conclusion: From ambition to advantage

ZS says the same thing about the KPI itself: pharma organizations have historically relied on "activity and adoption metrics that capture results only after they occur", treating activity as a proxy for impact (ZS, "From barriers to benchmarks: Redefining measurement in the new era of personalization", 2026). Tonight's engine

optimizes net incremental value, and reports suggestion acceptance and engagement as program KPIs, not outcomes.

Reflection A. “Aligning AI with business KPIs” — which KPI? Tonight's engine optimizes net incremental value (incremental NBRx × $1,200 − cost). Most omnichannel dashboards report engagement (opens, clicks, call completion). Write two sentences on what an engine that optimizes engagement would do to the action mix across the five Week 4 segments, and why (hint: the affinity matrix, notebook §4).

B. From call plans to orchestrated journeys — what an NBA engine is for (objectives 1 & 2)

The term is the industry's. Veeva introduced CRM Suggestions in 2015 as “data-driven recommendations on next best action and channel”. IQVIA's NBA fact sheet (2025): “an AI-powered recommendation engine … optimizing personal and digital customer engagement”, with “touchpoints … coordinated across channels and teams”. McKinsey (2022): “an optimized sequence of interactions for each HCP or segment to maximize the predicted sales return” — “the right HCP, with the right frequency, using the right channels”. EY lists a “next-best-action platform” among commercial-analytics components. ZS and Axtria now argue for a “next best experience”: ZS's critique is that “static, rules-based engines serve up repetitive, generic suggestions”; the answer is closed-loop learning that treats “every interaction as a learning event”.

Note the order in IQVIA's phrase “dynamic micro-segmentation and AI/ML-based scoring … 360-degree view of HCP preferences”: segmentation first, then preferences (affinity), then scoring. Tonight builds the middle piece as a recommender problem — affinity is observed for a subset of HCP × channel cells and predicted for the rest by collaborative filtering (Koren, Bell & Volinsky, 2009, in Part F), with the Week 4 micro-segments as the prior where there is no history.

IQVIA's 2026 paper on medical scientific engagement describes, for medical science liaisons, the loop an NBA engine automates for the commercial field: profile, prioritize, sequence touchpoints, log the outcome of each step, decide the next one. Read it as the business owner's description of the product.

> *High-quality engagement starts with robust customer profiling and flows into strategic orchestration of personal and non-personal channels. MSLs map influence networks, referral routes, formulary constraints, and evidence needs to prioritise stakeholders with pathway leverage. They then sequence touchpoints to sustain scientific momentum. For example, a focused virtual deep-dive is followed by a compliant, approved email summary and then a peer-to-peer roundtable to progress any evidence gap, with outcomes from each step logged to determine the next best action.*
>
> — IQVIA, “Redefining Global Medical Scientific Engagement” (2026), Insight-driven engagement: From call plans to orchestrated journeys

> *They balance in-person, virtual, and asynchronous touchpoints to protect quality without losing cadence.*
>
> — IQVIA, “Redefining Global Medical Scientific Engagement,” Performance and professionalism

And from IQVIA's 2025 paper on AI in Medical Affairs — the same idea from the content side: behavioral data → personalized communication → omnichannel delivery.

> *AI is also reshaping how Medical Affairs engages HCPs. By curating insights from behavioral data, AI can assist in the development of personalized communication strategies that reflect individual preferences and clinical interests. This digital engagement approach supports omnichannel delivery, whether through peer-*

> *reviewed articles, virtual symposia, or AI-powered chatbots, ensuring that scientific information reaches HCPs in the most effective format and context.*
>
> — IQVIA, "Harnessing the Power of Artificial Intelligence in Medical Affairs" (2025), Medical communications: Reshaping scientific dissemination

Note three things. First, the sequence in the first excerpt (deep-dive → approved email → roundtable) is a channel-sequencing rule — tonight's substitution finding (an email the week after a call is worth a fifth of a fresh one) is the commercial field's version of it, estimated rather than designed. Second, "with outcomes from each step logged" is the touch log; what the excerpt does not say is that the log must also record the probability with which each step was chosen, or nothing can be evaluated offline. Third, "formulary constraints" is in the profile — tonight's Access-constrained segment gets the access email, not the call, for exactly that reason.

Reflection B. Rewrite the MSL sequence in the first excerpt as an NBA policy: the context it needs, the eligibility rules it implies, and the one thing you would randomize in order to learn whether the roundtable step is worth its cost. Two short paragraphs.

C. The data layer an NBA engine stands on (objectives 2 & 3)

An NBA engine makes thousands of decisions a week from data that moves across CRM, email, web, access and Rx systems. IQVIA's 2026 piece on the agentic-era enterprise says what that requires — and names the failure mode of a bad decision engine in one sentence.

> *Agentic AI and other advanced technologies raise the stakes because they can interpret, recommend, orchestrate, and trigger action across workflows. But they cannot compensate for weak enterprise foundations. If data is fragmented, definitions are inconsistent, quality is questionable, and governance is weak, AI will not transform the enterprise. It will expose the disconnects leaders have not resolved.*
>
> — IQVIA, "AI Alone Will Not Modernize the Life Sciences Enterprise" (Adam Silvis, 2026), opening

> *In life sciences, data quality has always mattered. In the agentic era, it has become a front-line commercial issue. Poor data quality is no longer simply a reporting inconvenience or remediation task. It can translate into the wrong customer being prioritized, the wrong message being delivered, the wrong channel being selected, or the wrong signal being interpreted as an opportunity.*
>
> — IQVIA, "AI Alone Will Not Modernize the Life Sciences Enterprise," Bad data is not a back-office problem

> *As commercial teams move toward more personalized and dynamic engagement, the margin for ambiguity narrows. Omnichannel execution depends on the ability to connect touchpoints across field, digital, service, access, and engagement channels. If those touchpoints are not governed through a common data spine, teams may gain speed in isolated pockets while increasing fragmentation across the enterprise.*
>
> — IQVIA, "AI Alone Will Not Modernize the Life Sciences Enterprise," same section

> *The answer is not to slow innovation. It is to distinguish between what must be standardized and where teams can adapt and experiment. Core customer, product, territory, consent, compliance, and performance data require disciplined governance. Execution layers, including campaign design, content personalization, channel orchestration, and local experimentation, can then move faster because they are anchored to trusted foundations.*
>
> — IQVIA, "AI Alone Will Not Modernize the Life Sciences Enterprise," same section

> *The most durable use cases are embedded where decisions already happen: in customer relationship management systems, omnichannel planning tools, field preparation workflows, analytics environments,*

> *and mobile experiences that support customer-facing teams. Technology must reduce friction, not add another interface to manage.*
>
> — IQVIA, "AI Alone Will Not Modernize the Life Sciences Enterprise," Technology must be embedded in workflows

"The wrong customer being prioritized, the wrong message being delivered, the wrong channel being selected" is the response-model trap, the content rule and the affinity model — in one sentence. "Consent" is on the list of core data that requires disciplined governance: tonight's 2,450 non-consented emails and 135 complaints are what happens when it is not. "A common data spine" is the CDP. And "local experimentation" in the execution layer is the exploration arm — IQVIA puts it where tonight puts it, on the fast-moving side of the line.

Reflection C. Tonight's touch log has fourteen columns. Sort them into the paper's two tiers — core data that must be standardized, and execution-layer data that can move fast — and say which column, if it were wrong 5% of the time, would do the most damage to the NBA engine and why.

C2. Saturation, fatigue and frequency — the industry's own evidence (objective 3)

Tonight's fatigue curve is estimated on synthetic data; the phenomenon is documented in IQVIA's channel research. In "Channel Preference Versus Promotional Reality" (2018), the disconnect between what HCPs say they want and what they get "is much more common than convergence"; in the most digitally mature markets "the majority of HCP groups would prefer less communication via digital channels", and the initial success of digital "resulted in overkill and saturation". The specialties that receive the highest promotional spend per doctor are the ones that report it. IQVIA's Channel Dynamics work (2023) tracks the same thing under the name "digital fatigue", alongside stated channel preference shares (face-to-face 41%, in-person conferences 16%, email 10%). McKinsey's omnichannel model states the design goal as "the right HCP, with the right frequency, using the right channels, and with the right messaging". Frequency caps — a term borrowed from digital advertising — are how pharma orchestration encodes that evidence as a business rule; Axtria's framework names the three ingredients as "AI/ML models, business rules, and governance".

Reflection C2. IQVIA's evidence is about stated preference (what HCPs say in a survey); tonight's fatigue curve is about revealed response (what their prescribing did). Write three sentences on how the two can disagree, which one the frequency cap should be built on, and what the exploration arm adds that a preference survey cannot.

D. Patient predictive modeling and the governance that comes with it (objective 2)

The patient file is the Week 2/4 kind of asset the data vendors describe: Komodo Health's Healthcare Map holds longitudinal "patient journeys" — inpatient, professional and prescription claims — distinguishes "closed" patients, for whom the entire encounter history is captured, from "open" ones, and reads provider behavior (prescribing patterns, patient mix, adoption of new therapies) from those journeys. Tonight's §10 builds a discontinuation-risk model on the Week 4 patient file and turns it into a patient-side NBA. IQVIA's 2025 paper on trustworthy AI/ML for patient analytics describes what a model like that has to carry with it at scale: output validation, metrics at every stage, continuous monitoring, and a human in the loop.

> *Patient analytics and research, especially the secondary use of health information, is under increasing scrutiny from data protection authorities due to the use of sensitive health-related data sensitive and because the outputs will inform patient care and outcomes. Concerns may include fairness, transparency, and the risk of harm when personal health insights are used irresponsibly or without adequate safeguards.*
>
> — IQVIA, "Trustworthy AI/ML for Patient Analytics and Research" (2025), Introduction

> *Modeled outputs may take various forms, including risk stratification scores, predictive analytics, or cohort classifications. These outputs undergo a validation process to verify that they meet data protection standards, for example aggregation thresholds. Any external sharing is managed through secure, non-reversible pseudonymization methods, ensuring that the receiving systems cannot trace back to individual data points.*
>
> — IQVIA, "Trustworthy AI/ML for Patient Analytics and Research," AI-secure AI/ML

> *At the source ingestion and bridging stages, key metrics include outcome definition fidelity, fairness and bias detection, dimensionality compression validation, reconstruction risk measurement, and flagging for variable exclusions (such as geographic or sensitive attribute filters).*
>
> — IQVIA, "Trustworthy AI/ML for Patient Analytics and Research," Continuous monitoring of AI PrivOps metrics

> *During model training and inference, AI PrivOps tracks accuracy and loss curves, monitors drift and checks for abnormal fluctuations in model behavior through endpoint metrics. Variability (where applicable) is evaluated post-training, and motivated intruder testing may simulate adversarial scenarios. These techniques identify if outputs could leak sensitive patterns or invite misuse.*
>
> — IQVIA, "Trustworthy AI/ML for Patient Analytics and Research," same section

Map the excerpts onto tonight's six controls (Field Guide Part 8): "outcome definition fidelity" is the discontinuation label; "fairness and bias detection" is the calibration-by-payer check; "variable exclusions (such as geographic or sensitive attribute filters)" is the allowed-features list; "monitors drift" is the monthly re-check; "aggregation thresholds" and "non-reversible pseudonymization" are what happens to the score before it reaches the PSP. The paper describes a system that does these continuously; tonight's notebook does them once. The difference is the operating model, not the model.

Reflection D. Tonight's patient NBA sends 1,081 of the 1,200 highest-risk hold-out patients to "PSP enrollment outreach via the prescriber's office." Write the three-line consent and purpose statement that would have to be true for that outreach to be compliant, and name the one feature in the model you would most expect a privacy reviewer to challenge.

E. The syllabus reading and the PMSA items — a guided walk

E1. Athey & Imbens (2016), "Recursive Partitioning for Heterogeneous Causal Effects" — §1–3

What it is. The paper behind causal trees and (with Wager) causal forests: how to grow a tree that splits on the treatment effect rather than on the outcome, and why the training sample has to be split in two (one half to choose the partition, the other to estimate the effects in each leaf) for the leaf estimates to be honest.

- **§1 —** the problem statement: heterogeneous effects, no individual ground truth, and the warning that searching many subgroups and reporting the best is not inference. Read for the framing; it is tonight's §5b in one paragraph.
- **§2 —** the honest-estimation idea: the sample used to find the subgroups cannot be the sample used to estimate their effects. Tonight's split-by-HCP hold-out is the one-split version of this.

- **§3 —** the criterion: because τ(x) is unobserved, the paper derives an estimable proxy for the MSE of a partition (the 'EMSE'). Skim the derivation; the point is that a tree can be grown on it. Stop before §4 unless you want the simulation results.

What to take from it for tonight: the X-learner and the causal forest are two answers to the same question — how to estimate an effect that is never observed for anyone — and both pay for it with sample splitting and regularization. The Qini curve is the honest-estimation idea applied to evaluation.

E2. Veeva / Salesforce Marketing Cloud — NBA orchestration papers (syllabus: 'provided')

These are vendor documents and they describe the three layers of Part 2 from the inside: the CDP (identity + consent), the decisioning layer (rules + scores), the activation layer (CRM for the field, the email platform, digital). Links are in the readings file; they were not in the course library at the time of writing. Read them for the vocabulary (triggers, journeys, suppression, 'next best content'); read them with one question: where in the architecture is the exploration arm, and where is the logging propensity? If the answer is nowhere, the engine cannot be evaluated offline.

E3. The PMSA items

- **PMSA webinar — "Transforming Digital Pharma Engagement: Marketing-led, AI-driven Omnichannel Strategies."** Vendor-led; useful for the tension it exposes — a marketing-led NBA engine wants to orchestrate the rep, and the field wants the engine to suggest, not decide. Tonight's call-recency rule ('the field may override with a logged reason') is the compromise. Watch the first 20 minutes.
- **PMSA webinar (2 parts) — "From Multichannel to Omnichannel: How AI/ML Techniques Help Pharma Organizations Achieve Customer-Centricity."** The cleanest framing of the orchestration problem in the PMSA archive: from rules-based campaigns to ML-driven decisions. Part 1 is the framing (watch it); part 2 is the techniques (skim — it covers the response/uplift distinction and channel attribution).
- **PMSA 2025 conference session — "Operationalizing Next Best Action in Pharmaceutical Communications and Marketing."** Implementation-focused: signals, decisioning layer, activation, measurement. Tonight's §11 rule table is structured on it. Member archive; slides if you can get them.
- **Definitive Healthcare on PMSA 2025 — "3 ways PMSA 2025 showed us the future of pharma analytics."** Five-minute pre-read on the shift from retrospective dashboards to real-time decision support. Read it before Week 6 too — the forecasting theme is in it.

Reflection E. From the two-part webinar or the Veeva/Salesforce material: find one place where a vendor describes a 'propensity' or 'likelihood to engage' model as the engine's intelligence. Say in three sentences whether it is an affinity model or an uplift model, how you can tell, and what you would ask the vendor to randomize.

F. Industry sources for tonight's vocabulary

Every term used tonight is used the way these publications use it. The IQVIA items marked 'in the course library' are the PDFs in this folder; the rest are links (the Field Guide, Part 11, maps each term to its sources). Two items are not pharma publications and are included because they are the originating sources of a term: Radcliffe & Surry for uplift modelling and its four customer types, and the Li / Dudík papers for contextual bandits and offline policy evaluation.

- **IQVIA (2025).** IQVIA Next Best Action (fact sheet). NBA as an "AI-powered recommendation engine"; "touchpoints are coordinated across channels and teams"; "dynamic micro-segmentation and AI/ML-based scoring"; "360-degree view of HCP preferences"; tracks "acceptance of suggestions". — https://www.iqvia.com/-/media/iqvia/pdfs/library/fact-sheets/nba_fact_sheet_final1_2025.pdf

- **IQVIA (2022).** Outperform by 30% with Next Best: How AI is Revolutionizing Decision-Making in Commercial. “Behavior-changing intelligence within daily workflows”; “analytics-driven recommendations embedded within workflows”; “just-in-time actions”. — https://www.iqvia.com/library/white-papers/outperform-by-30-with-next-best
- **IQVIA (2024).** Time to Take Omnichannel Action: Delivering an Optimal Customer Experience Through the Harmony of People-Based and Digital Channels (Evans, Woods, Wooden, Bayley). “Orchestrated omnichannel harmony”; HCP “affinity”; “single customer view across all channels”; “next-best technology”; omnichannel scorecard. — https://www.iqvia.com/-/media/iqvia/pdfs/emea/library/whitepaper/time-to-take-omnichannel-action.pdf
- **IQVIA (2018).** Channel Preference Versus Promotional Reality: The Core Challenge of Multichannel Marketing (Dabbs, Rickwood, Smith, Wooden). Saturation: digital “overkill and saturation”; HCPs in the most-promoted specialties “would prefer less communication”; the gap between stated preference and promotional reality. — https://www.iqvia.com/-/media/iqvia/pdfs/library/white-papers/channel-preference-versus-promotional-reality.pdf
- **IQVIA (2023).** Choosing the right channel: HCP engagement preferences in a rapidly evolving environment (Channel Dynamics, T. Woods). “Digital fatigue”; channel-preference shares (F2F 41%, conferences 16%, email 10%). — https://www.iqvia.com/locations/emea/blogs/2024/10/choosing-the-right-channel-healthcare-professional-engagement
- **IQVIA (2026).** AI Alone Will Not Modernize the Life Sciences Enterprise (A. Silvis) — in the course library. “Common data spine”; consent as core governed data; “the wrong customer being prioritized, the wrong message being delivered, the wrong channel being selected”; “local experimentation” in the execution layer. — iqvia-info-solutions-leveraging-technology-article-2026.pdf
- **IQVIA (2026).** Redefining Global Medical Scientific Engagement — in the course library. “From call plans to orchestrated journeys”; “outcomes from each step logged to determine the next best action”; “orchestration of personal and non-personal channels”. — hcp-engagement-medical-affairs-white-paper.pdf
- **IQVIA (2025).** Trustworthy AI/ML for Patient Analytics and Research — in the course library. Risk stratification scores and their validation; fairness and bias detection; drift monitoring; human-in-the-loop governance for patient analytics. — trustworthy-aiml-for-patient-analytics-and-research---final.pdf
- **ZS (2026).** When recommendations aren't enough: Rethinking HCP engagement through next best experience (R. P. Lal et al.). “Static, rules-based engines serve up repetitive, generic suggestions”; next best experience; “closed-loop learning” — “every interaction as a learning event”; “agentic AI with human-in-the-loop controls”. — https://medium.com/zs-associates/when-recommendations-arent-enough-rethinking-hcp-engagement-through-next-best-experience-bc6c2d0904ae
- **ZS (2026).** Synthetic test and control: Pharma impact measurement (A. Patel et al., with GSK). Test-and-control as “the backbone of success measurement”; holdout groups; synthetic control when a preplanned holdout is not available. — https://www.zs.com/insights/synthetic-test-and-control-in-pharma-impact-measurement
- **ZS (2026).** From barriers to benchmarks: Redefining measurement in the new era of personalization (Powers, Cohen, Jackson). “Activity and adoption metrics” as a proxy for impact — why engagement is not the outcome. — https://www.zs.com/insights/barriers-to-benchmarks-pharma-measurement-engagement
- **ZS (2024).** 5 ways ZAIDYN OmniBERT boosts next best action in pharma marketing. Signals and “moments that matter” (a diagnosis, a formulary change); NBA “suggestions”; “sales lift”. — https://zaidyn.zs.com/blogs/next-best-action-for-pharma-marketing-teams
- **Axtria (2025).** A Customer-Centric Omnichannel Orchestration Framework for Optimizing Promotion Efforts (with Bausch + Lomb). “AI/ML models, business rules, and governance combine to drive precision targeting”; “test-and-learn cycles”; NBA-driven engagement. — https://www.axtria.com/reports/a-customer-centric-omnichannel-orchestration-framework-for-optimizing-promotion-efforts
- **Axtria (2025).** The Future of Omnichannel Pharma Sales: Moving Beyond NBA to Drive HCP Engagement (P. S. Bhatia). NBA as “channel- and content-driven suggestions”; next best experience; “suggestion explainability” for rep trust; MMM + attribution as unified measurement. — https://www.axtria.com/articles/the-future-of-omnichannel-pharma-sales-moving-beyond-nba-to-drive-hcp-engagement
- **Axtria (2021).** Promotion Response Modeling for Pharma (PMSA webinar, D. Wood & R. Kumar). “Promotion response” — the impact of promotional activity on prescribing; diminishing returns; promotion across multiple channels. — https://www.axtria.com/blog/promotion-response-modeling
- **Axtria (2020).** Axtria Triggers for Veeva CRM Suggestions (data sheet). Triggers → suggestions delivered to the rep in CRM. — http://insights.axtria.com/hubfs/Datasheets/Axtria_Triggers_Data_Sheet.pdf

- **McKinsey (2022).** Demystifying the omnichannel commercial model for pharma companies in Asia (Chan, Kim, Le Deu, Li). “An optimized sequence of interactions for each HCP or segment to maximize the predicted sales return”; “360-degree view of HCPs”; “the right HCP, with the right frequency, using the right channels”; “A/B testing in the field”; reps feed outcomes back. — https://www.mckinsey.com/jp/en/our-insights/demystifying-the-omnichannel-commercial-model-for-pharma-companies-in-asia
- **McKinsey (2025).** Reimagining life science enterprises with agentic AI (Adabala Viswa et al.). “Orchestrating agents”; “agents must consult humans before making important decisions”; guardrails set by the enterprise. — https://www.mckinsey.com/industries/life-sciences/our-insights/reimagining-life-science-enterprises-with-agentic-ai
- **EY (2023).** Omnichannel customer experience (Life Sciences). “Next-best-action platform” as a commercial-analytics component; “omnichannel orchestration”; breaking data silos across sales, marketing and medical. — https://www.ey.com/en_us/industries/life-sciences/omnichannel-customer-experience
- **Komodo Health (2023).** A More Complete View of Provider Behavior (B. Woywod). Longitudinal “patient journeys”; “closed” vs “open” patients; provider behavior read from patient-level claims. — https://www.komodohealth.com/perspectives/a-more-complete-view-of-provider-behavior/
- **Veeva Systems (2015).** Veeva CRM Suggestions — data-driven recommendations on next best action and channel (press release). “Suggestions” as the field-facing form of NBA; “next best action and channel”. — https://s206.q4cdn.com/200001835/files/doc_news/2015/12/1/New-Veeva-CRM-Suggestions-Now-Available-Powers-Sales-Teams-with-Data-Driven-Recommendations-on-Next-Best-Action-and-Channel.pdf
- **Stochastic Solutions (N. Radcliffe & P. Surry) (2011).** Real-World Uplift Modelling with Significance-Based Uplift Trees (white paper). Uplift modelling; the four customer types — persuadables, sure things, lost causes, sleeping dogs. The originating industry source (not pharma-specific). — https://stochasticsolutions.com/pdf/sig-based-up-trees.pdf
- **Athey & Imbens, PNAS (2016).** Recursive Partitioning for Heterogeneous Causal Effects. Causal trees / honest estimation — the syllabus reading behind causal forests. — https://www.pnas.org/doi/10.1073/pnas.1510489113
- **Koren, Bell & Volinsky, IEEE Computer (2009).** Matrix Factorization Techniques for Recommender Systems. Collaborative filtering by matrix factorization with biases — the method tonight uses to predict channel affinity for HCPs and channels never observed together. Not a pharma source. — https://datajobs.com/data-science-repo/Recommender-Systems-[Netflix].pdf
- **Li, Chu, Langford & Schapire (WWW); Dudík, Langford & Li (ICML) (2010).** A Contextual-Bandit Approach to Personalized News Article Recommendation (2010); Doubly Robust Policy Evaluation and Learning (2011). Contextual bandits, offline policy evaluation (IPS, doubly robust). From the online-advertising literature — no pharma brand uses these words; tonight labels them as such. — https://arxiv.org/abs/1003.0146 · https://arxiv.org/abs/1103.4601

Bring to Week 6

One written answer (any of A–E) and your Part B rule table from the exercise. Week 6 forecasts the brand you have now orchestrated — and Homework 2 is released.